Power Sets :-

$\rightarrow \quad A = \{1, 2, 3\} \rightarrow 2^{|A|}$

$\mathcal{P}(A) = \{\{\}, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\}\}$

power set of A

$\boxed{\mathcal{P}(A) = \{X : X \subseteq A\}} \rightarrow$ mathematical def.

$|\mathcal{P}(A)| = 2^{|A|} \rightarrow$ number of subsets of A.

ex:-

$\mathcal{P}(\{0,3\}) = \{\{\}, \{0\}, \{3\}, \{0,3\}\}$

$|\mathcal{P}(\{0,3\})| = 2^2 = 4$

ex:- $\mathcal{P}(\emptyset) = ? = \{\{\}\} = \{\emptyset\}$

$|\mathcal{P}(\emptyset)| = 2^{|\emptyset|}$

$= 2^0$

$= 1$

$|\{\emptyset\}| = 1$ ✓

$\cancel{|\emptyset|} = 0$

ex:- $\mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$

$2^1 = 2$

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$\boxed{\mathcal{P}(1)} = \{\emptyset, \{1\}\}$

True or false ?

Wrong notation

ex:- $\mathcal{P}(\{a\}) \times \mathcal{P}(\{\emptyset\}) = ?$

$\{\emptyset, \{a\}\} \times \{\emptyset, \{\emptyset\}\}$

$\{(\emptyset, \emptyset), (\emptyset, \{\emptyset\}), (\{a\}, \emptyset), (\{a\}, \{\emptyset\})\}$

ex:- $\mathcal{P}(\mathbb{N}) = ?$

$\mathbb{N} \rightarrow$ infinite

$|\mathcal{P}(\mathbb{R})| = ? \quad 2^{|\mathbb{R}|} \Rightarrow 2^{\infty} = \infty$